



Yunsu Jang

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Education

AMS Institute - Joint Degree TU Delft, Wageningen University

M.Sc. IN METROPOLITAN ANALYSIS, DESIGN AND ENGINEERING (MADE)

Amsterdam, Netherlands

Sep. 2025 - Aug. 2027 (Expected)

Korea University

B.S. IN CIVIL, ENVIRONMENTAL AND ARCHITECTURAL ENGINEERING

Seoul, S. Korea

Mar. 2019 - Feb. 2025

Technical University of Berlin

B.S. IN CIVIL ENGINEERING

Berlin, Germany

Aug. 2023 - Aug. 2024

- Took three project courses related to building systems in the M.Sc. track in Civil Systems Engineering.

Working Experience

P3 Korea

Seoul, S.Korea

INTERN

Mar. 2025 - Jul. 2025

- Joined the electric vehicle(EV) Interoperability(IOP) test team in its formative stage, designed the complete test framework, planned drive-route scenarios, and authored the final compatibility report for pre-launch EVs
- Spearheaded market research on EVs and charging infrastructure, mapping domestic and global ecosystems and building a comprehensive Excel- and PowerPoint-based dashboard to synthesize findings
- Contributed to battery-technology consulting projects by conducting global market analyses and co-developing a cost-modeling tool that benchmarked competitors and quantified component-level cost gaps

Siemens Energy

Kirchheim unter Tech, Germany

WORKING STUDENT

May. 2024 - Aug. 2024

- Engaged with the sales team in a leading transformer manufacturing facility, enhancing client relationships through strategic communication and support
- Employed AutoCAD to revise and tailor documentation in alignment with client specifications
- Facilitated cross-departmental coordination with the transformer casing manufacturing corporation to effectively convey client requirements and secure necessary modifications
- Has a deep understanding of transformers, which serves as a foundation for grasping how energy systems operate and for establishing energy efficiency strategies and solutions at the city level
- Actively organized client meetings, providing insights and constructive feedback for project enhancements

Hyundai Engineering and Construction

Seoul, S.Korea

INTERN

Jan. 2023 - Feb. 2023

- Took charge of the foundation phase of a Pedestrian bridge and Ramp construction project, gaining firsthand experience in construction methodologies
- Provided insight in major construction activities, especially including the practical application of construction processes
- Actively engaged in on-site experiments, such as the Concrete Compressive Strength Test and Rock Identification Test, which were instrumental in evaluating material performance

Extracurricular Activity

Career Committee

Amsterdam, Netherlands

- Conducted company research for the career market and managed cold outreach, including follow-up communications
- Planned, structured, and coordinated the organization of the career market

Sep. 2025 - Present

Korea University Scuba Diving Club

S.Korea

- Organized and facilitated seasonal scuba diving expeditions, providing education based on Physics *Mar. 2019 - Feb. 2025*
- Conducted a detailed verification of the layout using AutoCAD and SketchUp throughout the remodeling process of the club room
- Delivered comprehensive instruction in scuba diving to junior members, serving as a dive buddy during practical expeditions to ensure adherence to safety protocols
- Spearheaded promotional campaigns through the university's application, raising awareness and engagement within the diving community

Hillstate Dream Mentoring Volunteers

Seoul, S.Korea

- Acted as a university head mentor in a collaborative team, providing guidance and support to high school mentees *Sep. 2022 - Aug. 2023*
- Led monthly career experience programs and lectures, structuring mentees' perspectives on career paths
- Conducted weekly mentoring sessions focused on Mathematics and Physics, fostering academic growth and critical thinking skills in mentees
- Supervised fellow university mentors as a leader to refine mentoring strategies, ensuring a more effective and impactful mentoring experience

Korea Army

Paju, S.Korea

- Engaged in tactical training programs at the Korea Combat Training Center (KCTC) *Mar. 2021 - Sep. 2022*
- Served in a reconnaissance company, conducting mine detection and ambush operations within the Demilitarized Zone (DMZ), showcasing adaptability and strategic planning in high-pressure situations

Engineering College Student Council

Seoul, S.Korea

- Planned and managed the "Engineering League," an inter-departmental soccer tournament *Mar. 2019 - Feb. 2020*
- Organized and executed the Engineering College Fall Festival, overseeing all event logistics and programming to ensure a successful and engaging experience for attendees

Skills

Program	Auto CAD, Microsoft Office, Dynamo, Open Studio, Energy Plus
Programming	R, Python
Language	Korean (Native), English (Fluent), German (B1), Dutch (Basic)
Driver License	Category B (Netherlands)

Project

Construction Comprehensive Design (Water and Environment)

Seoul, S.Korea

- Investigated optimal sites for deploying early tsunami detection systems at Hanul Nuclear Power Plant, aiming to enhance disaster preparedness and response strategies in that region *WS, 2024/25*
- Executed a collaborative project utilizing the COMCOT simulation tool to model and analyze hydrodynamic behaviors

Whole Life Civil Systems Analysis

Berlin, Germany

- Conducted LCA (Life Cycle Analysis) of HVAC systems utilizing R programming *WS, 2023/24*
- Collaborated with a team of four to synthesize individual systems, performing a comprehensive LCA for a nearshore wind farm, contributing to sustainable design practices

Multi-Physics approaches for modeling civil systems

Berlin, Germany

- Addressed complex engineering challenges and performed detailed model analyses using advanced software platforms, EnergyPlus and Frame3DD *WS, 2023/24*
- Led a team of five to analyze and improve energy and humidity control measures for Berlin's Martin-Gropius-Bau building using OpenStudio

Modeling Civil Engineered Systems

Berlin, Germany

- Developed HVAC system ontologies using Protégé software, fostering enhanced interoperability and design efficiency *WS, 2023/24*
- Designed and simulated shading devices, calculating their effects on energy savings using Dynamo, to improve thermal comfort and efficiency
- Collaborated on the "Water Supply for Urban Areas" modeling project, integrating ontologies and parametric design methodologies for holistic urban development

Building Energy Modeling and Analysis

Seoul, Korea

- Created a detailed simulation model in MATLAB to evaluate the energy performance of residential spaces, analyzing various performance parameters and optimization strategies *WS, 2022/23*
- Enhanced energy efficiency by integrating a Horizontal Ground Heat Exchange Ventilation System and a Heat Exchanger, validating design enhancements through rigorous numerical analyses